

Seeing Through Deepfakes

A Detection System Using Transparency Signals in Generative AI

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Abstract

The rapid spread of generative AI has enabled the large-scale circulation of highly realistic synthetic media, raising significant risks of misinformation, manipulation and erosion of trust. While the EU AI Act and the Draft Code of Practice on Transparency introduce obligations for identifying AI-generated content, practical and robust implementation remains unclear. This research proposes a proof-of-concept deepfake detection system that integrates multiple transparency signals and model-based visual classification. By benchmarking existing approaches and evaluating a multi-signal framework under realistic conditions, the study aims to assess the effectiveness, robustness and regulatory relevance of transparency based detection mechanisms, supporting trustworthy identification of AI-generated and manipulated imagery.

1. Introduction

1.1 Defining the Problem

The rapid diffusion of generative AI systems has enabled the large-scale creation and dissemination of highly realistic synthetic media, including images, video, text and audio. While regulatory efforts within the European Union (EU) increasingly emphasise transparency obligations for AI-generated content, the global and decentralised nature of digital platforms means that such content can be created, modified and shared across jurisdictions beyond the EU's regulatory reach. As a result, AI-

generated and manipulated media originating from third countries can circulate freely within the Union, undermining transparency safeguards and increasing the risk of misinformation, deception and malicious manipulation.

Existing deepfake detection approaches remain fragmented, often relying on single technical signals that are vulnerable to evasion or degradation. There is therefore a pressing need for robust detection mechanisms that can identify AI-generated content through multiple transparency signals, supporting both regulatory enforcement and informed decision-making. Addressing this gap is essential to mitigating the societal and democratic harms associated with the unchecked spread of synthetic media.

1.2 Motivation

The motivation for this research is grounded in recent regulatory developments at EU level, particularly the publication of the first draft Code of Practice (CoP) on Transparency for AI-Generated Content by the EU AI Office. The draft CoP operationalises the transparency obligations set out in Article 50 of the EU AI Act by proposing concrete technical measures to enable the identification and verification of AI-generated or manipulated content [1], [2]. However, while the CoP outlines what transparency outcomes are expected, it remains largely agnostic as to how these measures should be combined, validated, or assessed in practice, especially under adversarial or cross-platform conditions. This creates uncertainty

for both providers and regulators regarding the effectiveness and reliability of proposed transparency mechanisms.

This research is motivated by the need to empirically examine and operationalise these transparency signals within a coherent detection framework, contributing practical insight into how the CoP can be meaningfully implemented and evaluated in real-world contexts.

1.3 Aim & Objectives

The aim of this research is to design and evaluate a proof-of-concept deepfake detection system that integrates multiple transparency signals to support the identification of AI-generated content in line with emerging EU transparency requirements.

To achieve this, the following objectives are to be met:

1. To research on the regulatory and technical taxonomy of transparency signals for AI-generated content, based on the EU AI Act and the draft CoP.
2. To evaluate existing deepfake detection and provenance approaches with respect to robustness, generalisability and alignment with regulatory transparency expectations.
3. To design a proof-of-concept detection system that integrates metadata analysis, watermark detection and model-based visual classification within a unified framework.
4. To develop a conditional deepfake assessment mechanism that distinguishes harmless AI-generated imagery from content involving semantic resemblance to real persons, objects, places, or events.
5. To assess system performance comparatively, using established benchmarks and survey of expert feedback to evaluate effectiveness against existing detection approaches.

2. Background

2.1 Limits of Deepfake Detection

Synthetic content refers to media that is generated or significantly altered by algorithmic systems [3]. Deepfakes represent a subset of such content, characterised by highly realistic AI-generated or manipulated images, audio, video or text that falsely appear authentic [1]. Existing detection approaches primarily rely on convolutional neural networks to identify pixel-level artefacts, alongside multimodal methods exploiting audio-visual inconsistencies [4]. Forensic analysis further examines cues such as physiological irregularities, lighting inconsistencies, and metadata anomalies [5]. Despite advances, detection systems exhibit limited generalisation to unseen datasets and emerging generative techniques. Their robustness is further undermined by common post-processing operations, including compression, and by adversarial perturbations designed to evade classifiers, highlighting the inherent fragility of purely content-based detection approaches [4], [5].

2.2 Transparency Signals

Transparency signals encompass proactive mechanisms for identifying AI-generated or manipulated content. Watermarking embeds identifiable signals within pixel space or latent representations, while cryptographic provenance frameworks such as Coalition for Content Provenance and Authenticity (C2PA) record content history through tamper-evident, signed manifests ("Content Credentials"). Metadata-based approaches further encode identity and integrity assertions using standardised schemas [6]. Unlike forensic detection, which analyses completed content for statistical artefacts, in-generation and provenance-based methods integrate traceability during synthesis or bind verifiable records to content at creation [7]. However, these approaches face challenges related to robustness against removal and forgery attacks, as well as interoperability across varied platforms and distribution environments.

2.3 Explainability & Human Trust

Explainable detection frameworks, such as those employing Grad-CAM, enhance trust by localising features that contribute to classification decisions, including facial distortions or physiological inconsistencies, thereby supporting human interpretability [4]. Hybrid multimodal approaches further improve reliability by combining model outputs with forensic cues and metadata-based assertions [5]. However, explainability alone is insufficient if signals remain inaccessible to non-expert users.

To address risks of misinformation and epistemic doubt, transparency mechanisms must be both verifiable and meaningful. Within the EU regulatory context, the AI Act and the Draft Code of Practice on the transparency of AI-Generated Content emphasise meaningful disclosure obligations, including human-understandable indicators and standardised, interactive transparency cues [1], [2]. Content credentials thus move beyond binary labels, offering contextualised information about an asset's creation and modification history [3].

2.4 Legislative & Regulatory Gaps

Article 50 (2) of the EU AI Act requires that generative AI outputs be detectable and, where applicable, identifiable through machine-readable means [1]. The Code of Practice operationalises this obligation through layered transparency signals, including watermarking and metadata-based provenance. It further envisages shared or aggregated detection verifiers and obliges providers to enable third-party verification through freely accessible detection interfaces [2].

However, a significant technical gap remains. Existing transparency mechanisms are fragmented, inconsistently implemented and susceptible to adversarial circumvention. While the CoP articulates desired transparency outcomes, it remains largely agnostic on how such measures should be empirically validated across real-world, cross-platform and jurisdictionally diverse deployment environments.

3. Proposed Methodology

3.1 Regulatory & Technical Mapping

This phase will establish a structured mapping between regulatory transparency obligations and corresponding technical signals for AI-generated content. Relevant provisions of the EU AI Act and the Draft Code of Practice shall be analysed to derive concrete technical requirements. These requirements will be systematically aligned with transparency mechanisms resulting in a taxonomy that links legal expectations to implementable verification approaches.

3.2 Benchmarking Existing Models

This phase benchmarks representative deepfake detection and transparency verification models across metadata analysis, watermark detection and content-based classifiers. Selected models will be evaluated using standard datasets and metrics to assess accuracy, robustness and generalisation. The analysis will identify performance gaps, particularly under adversarial or cross-platform conditions, informing the design choices and limitations addressed in the proposed proof-of-concept system.

3.3 Multi-Signal Classification

This phase develops a modular proof-of-concept system that operationalises regulatory transparency signals for assessing AI-generated imagery. The system initially analyses content using three independent components identified in the first Draft CoP:

1. metadata analysis,
2. watermark detection, and
3. model-based visual classification.

Each component produces a probabilistic estimate indicating whether the content is AI-generated. These outputs are subsequently normalised and aggregated into a single confidence score, reflecting the overall likelihood of synthetic origin. This multi-signal design enables comparative evaluation of heterogeneous transparency mechanisms while mitigating over-reliance on any single detection method. The modular architecture supports

extensibility, transparency of system reasoning, and alignment with regulatory expectations concerning explainability, robustness, and proportional risk assessment in AI-generated content verification.

3.4 Deepfake Assessment

Where the aggregated confidence score exceeds a predefined threshold indicating likely AI-generated content, the system initiates a secondary semantic analysis stage. This stage examines whether the generated content resembles existing persons, objects, places, entities or events, consistent with regulatory definitions of deepfakes [1]. This step determines if AI-generated content is likely a deepfake rather than harmless synthetic imagery. This assessment would focus on semantic resemblance rather than solely visual artefacts, reflecting concerns around impersonation, misinformation and identity misuse. By conditioning deeper analysis on prior AI-generation likelihood, the approach remains proportionate and computationally efficient.

3.5 Validating Performance

This phase evaluates the proposed system using quantitative benchmarking methods consistent with those researched and applied in Section 3.2. The integrated framework will be evaluated on chosen datasets using common metrics like accuracy, precision, recall, and by testing its robustness in both normal and adversarial settings. Performance will be compared against baseline models to evaluate whether the multi-signal aggregation improves reliability and reduces false positives, particularly in cross-platform or degraded-content scenarios.

In parallel, a qualitative evaluation will be conducted through an expert survey, following an established methodological approach previously applied in similar evaluations [8]. Domain experts in AI, digital forensics and regulation will assess the system's usefulness, interpretability, and trustworthiness. Where appropriate, an exploratory survey with non-expert participants may be considered to assess whether the system aids individuals, without AI expertise, in recognising AI-generated or

manipulated content. This mixed-method evaluation supports both technical validity and human-centred regulatory relevance.

4. Expected Outcomes

4.1 Transparency Signal Mapping

The transparency signal mapping is expected to yield a clear, regulation-aligned taxonomy linking legal transparency obligations to verifiable technical indicators of AI-generated content. By systematically translating requirements from the EU AI Act and the Draft Code of Practice into observable signals, this mapping will clarify which forms of evidence are technically actionable and where gaps or overlaps exist. The resulting taxonomy will support consistent interpretation of regulatory expectations, inform detector design choices and provide a structured reference for evaluating how well technical transparency mechanisms align with emerging governance standards.

4.2 Integrated Framework

The integrated framework is expected to demonstrate that combining heterogeneous transparency signals produces more reliable identification of AI-generated content than single signal approaches. By structuring detection in sequential stages, the framework will enable clearer separation between general AI generation and high-risk deepfake content involving people or objects. An important outcome is improved robustness to partial signal failure and adversarial evasion, alongside more interpretable confidence outputs. Collectively, these outcomes will illustrate the practical value of multi-signal integration for scalable, regulation-aligned detection of AI-generated and manipulated imagery.

4.3 Comparative Performance

The comparative performance analysis is expected to demonstrate that the proposed multi-signal approach achieves more stable and robust detection outcomes than existing single-method techniques. By benchmarking against

representative deepfake detectors and provenance-based systems, the evaluation will highlight relative strengths in accuracy, generalisability and resilience to signal degradation or adversarial manipulation. Rather than optimising raw performance alone, the outcome will emphasise consistency across scenarios and alignment with transparency objectives. These results will provide empirical support for the added value of integrating multiple transparency signals within a single detection pipeline.

4.4 Practical Limitations

The analysis is expected to highlight several practical limitations affecting real world deployment of transparency based detection. Existing research indicates that screenshots, platform-applied filters, recompression and cropping can degrade or entirely remove certain transparency signals, especially metadata and fragile watermarks. As a result, detection reliability may vary depending on content provenance and distribution pathways. These limitations underscore the difficulty of enforcing transparency guarantees once content leaves controlled generation environments. Acknowledging such constraints will help contextualise performance results and clarify the realistic scope of transparency based deepfake detection under current conditions.

4.5 Scalability & Future Research

Future work is expected to explore how the proposed approach could scale beyond individual detectors towards a coordinated, interoperable verification mechanism. One potential direction is the development of a unified or ‘federated’ detection layer, conceptually aligned with a union-level transparency mark or verifier, capable of aggregating signals across providers and platforms. Such an approach could support consistent public facing indicators, enabling EU citizens to more easily assess whether content is AI-generated or constitutes a deepfake. Further research is needed to address governance, standardisation and trust assumptions underpinning such large-scale deployment.

5. Conclusion

This proposal has outlined a research agenda addressing the growing gap between regulatory transparency obligations for generative AI and the technical mechanisms available to operationalise them in practice. As AI-generated and manipulated visual content becomes increasingly indistinguishable from authentic media, existing disclosure approaches remain fragmented, fragile and poorly aligned with regulatory intent. In response, this thesis proposes a structured investigation into how multiple transparency signals can be systematically mapped, evaluated and integrated to support trustworthy identification of AI-generated images and deepfakes.

The proposed methodology adopts a regulatory first perspective, grounding technical design choices in the transparency requirements articulated within European governance instruments. By benchmarking existing detection and provenance techniques against these requirements, the research aims to provide empirical insight into their respective strengths, limitations and failure modes under realistic conditions, including post processing, screenshots, and platform level transformations. Building on this analysis, the dissertation advances a conceptual multi-signal detection framework that combines provenance cues, watermarking signals and forensic deepfake indicators, with the objective of improving robustness and interpretability beyond what single-method approaches can offer.

Importantly, this work does not seek to claim definitive detection accuracy or to propose a production-ready enforcement tool. Rather, its contribution lies in clarifying how transparency mechanisms can be meaningfully evaluated as regulatory instruments, highlighting trade-offs between technical resilience, usability and scalability. The anticipated outcomes include a clearer understanding of where current transparency measures succeed or fail, and how integrated verification approaches may better support regulators, platforms and end-users in assessing the authenticity of visual content.

Finally, the dissertation positions itself as a foundation for future research into interoperable, union level verification infrastructures capable of supporting shared detection and disclosure mechanisms across jurisdictions. By bridging legal expectations and technical feasibility, this research aims to contribute to ongoing policy discussions on AI transparency while offering a principled basis for the development of more robust, accountable and societally aligned deepfake detection systems.

References

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